Lab No: 01

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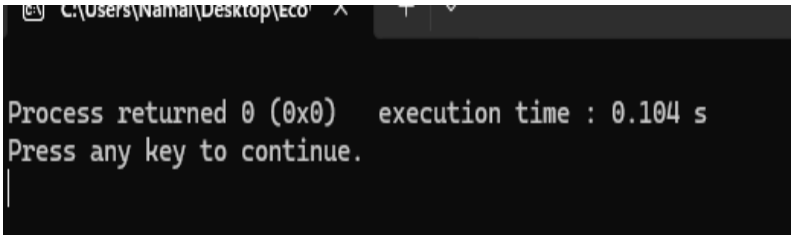
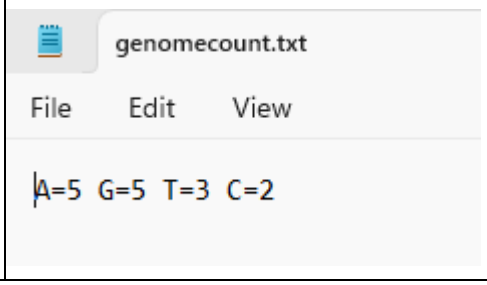
Roll No: 17

Task 1:	

Code:	#include <iostream>
	#include <fstream>
	using namespace std;
	
	int main() {
	  ifstream infile("GENOME.txt");
	  ofstream outfile("genomecount.txt");
	
	  int a = 0, g = 0, t = 0, c = 0;
	  char ch;
	
	  while (infile >> ch) {
	  if (ch == 'A') a++;
	  else if (ch == 'G') g++;
	  else if (ch == 'T') t++;
	  else if (ch == 'C') c++;
	  }
	
	  outfile << "A=" << a << " G=" << g << " T=" << t << " C=" << c;
	
	  infile.close();
	  outfile.close();
	  return 0;
	}

Screenshots:



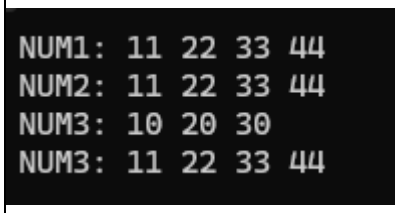


Task 2:

Code:

```
#include <iostream>
using namespace std;

class Numbers
{
private:
   int s;
   int *a;

public:

   Numbers(int x, int y[])
   {
      s = x;
      a = new int[s];
      for (int i = 0; i < s; i++)
      {
         a[i] = y[i];
      }
   }

   void data()
   {
      for (int i = 0; i < s; i++)
      {
         cout << a[i] << " ";
      }
      cout << endl;
```

```
    }

    Numbers(const Numbers &obj)
    {
        s = obj.s;
        a = new int[s];
        for (int i = 0; i < s; i++)
        {
            a[i] = obj.a[i];
        }
    }

    Numbers& operator=(const Numbers &obj)
    {
        if (this != &obj)
        {
            delete[] a;
            s = obj.s;
            a = new int[s];
            for (int i = 0; i < s; i++)
            {
                a[i] = obj.a[i];
            }
        }
        return *this;
    }

    ~Numbers()
    {
        delete[] a;
    }
};

int main()
{
    int arr[4] = {11, 22, 33, 44};

    Numbers NUM1(4, arr);
    cout << "NUM1: ";
    NUM1.data();


    Numbers NUM2=NUM1;
    cout << "NUM2: ";
    NUM2.data();
```

```
    int arr2[3] = {10, 20, 30};
    Numbers NUM3(3, arr2);
    cout << "NUM3: ";
    NUM3.data();

    NUM3 = NUM1;
    cout << "NUM3: ";
    NUM3.data();


    return 0;
}
```

Screenshot: